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Introspection et Débogage avec pal_statistics dans ROS 2 Control

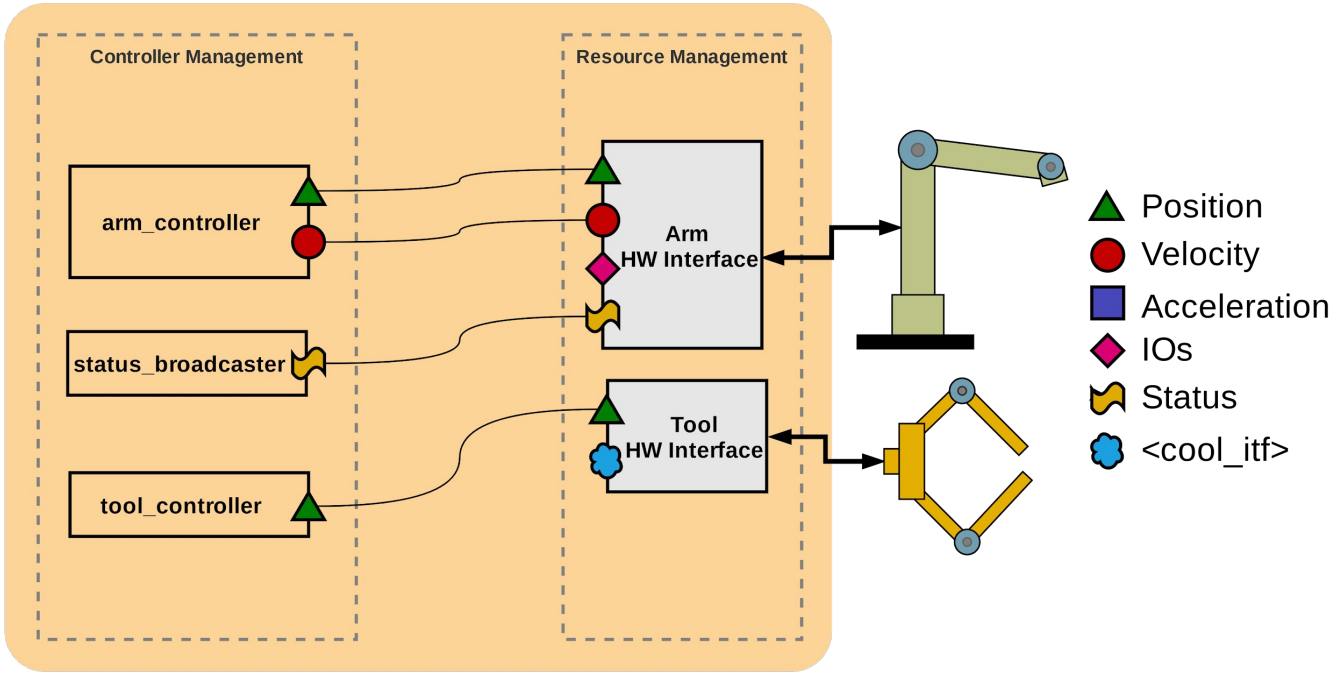
Maximilien Naveau et Pierre Fernbach | November 2025

Introduction

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Control Temps Réel et ROS2 control

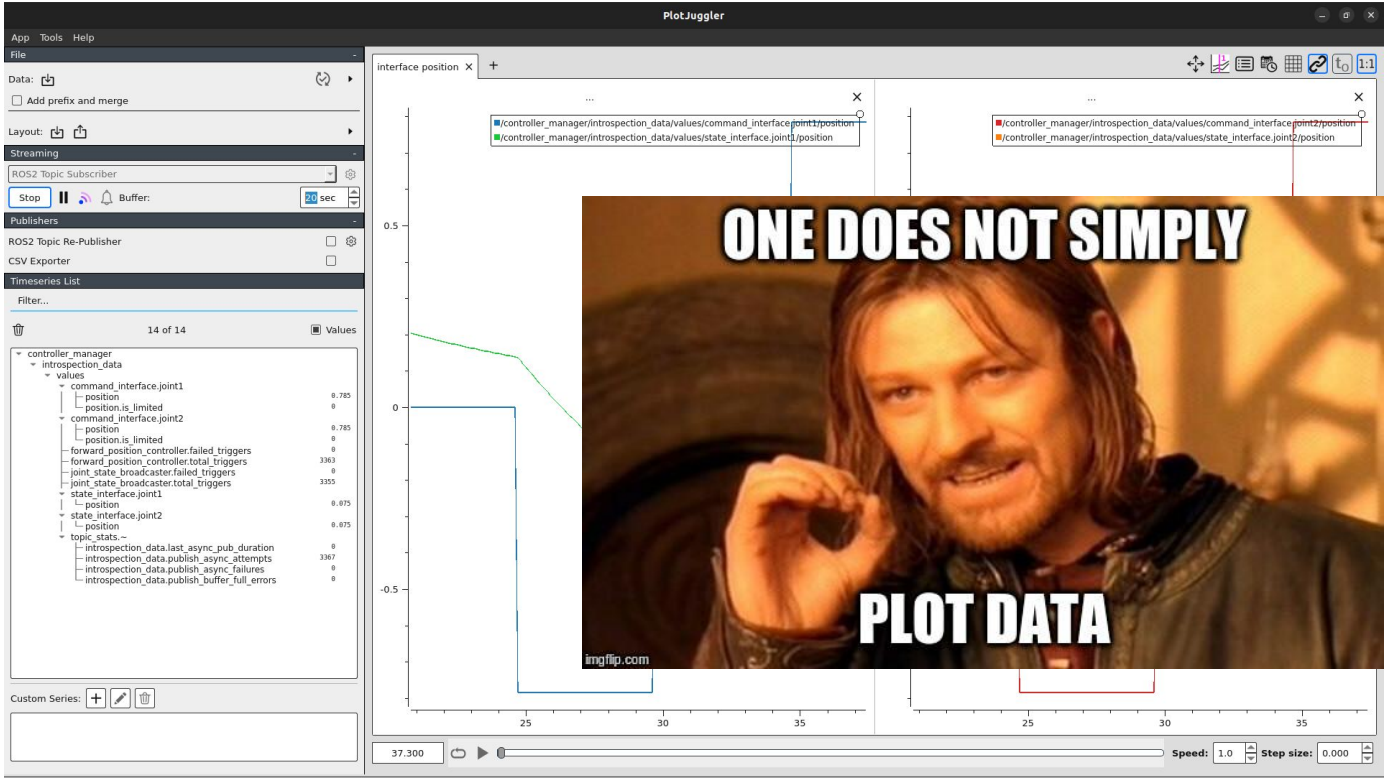
https://control.ros.org/jazzy/doc/resources/resources.html



CC-BY: Denis Stogl, Bence Magyar (ros2_control)

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PlotJuggler



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pal_statistics

https://github.com/pal-robotics/pal_statistics

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- Real time software
- ROS2 compatible
- Based on ROS2 topics
- Stream data efficiently
- Easy to use
- C++/Python
- Open-source software

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C++ snippet

```cpp
#include <pal_statistics/pal_statistics_macros.h>

void do_monitor_temperature() {
  double temperature;
  pal_statistics::RegistrationsRAII registrations;
  REGISTER_VARIABLE("/statistics_topic", "motor temperature", &temperature, &registrations);
  while (ros::ok()) {
    temperature = read_temperature();
    ros::Duration(0.1).sleep();
    PUBLISH_STATISTICS("/statistics_topic");
  }
  UNREGISTER_VARIABLE("/statistics_topic", "motor temperature");
}
```

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Python snippet

```
import rclpy

from pal_statistics.statistics_registry import StatisticsRegistry

def do_monitor_temperature():
  node = rclpy.node.Node()
  temperature = 0.0;
  registrations = StatisticsRegistry("/statistics_topic", node)
  registrations.registerFunction("motor temperature", lambda: temperature)
  while rclpy.ok():
    temperature = read_temperature();
    node.get_clock().sleep_for(rclpy.duration.Duration(seconds=0.1))
    registrations.publish()
```

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pal_statistics in ROS2 control

Pal_statistics in ROS2 control

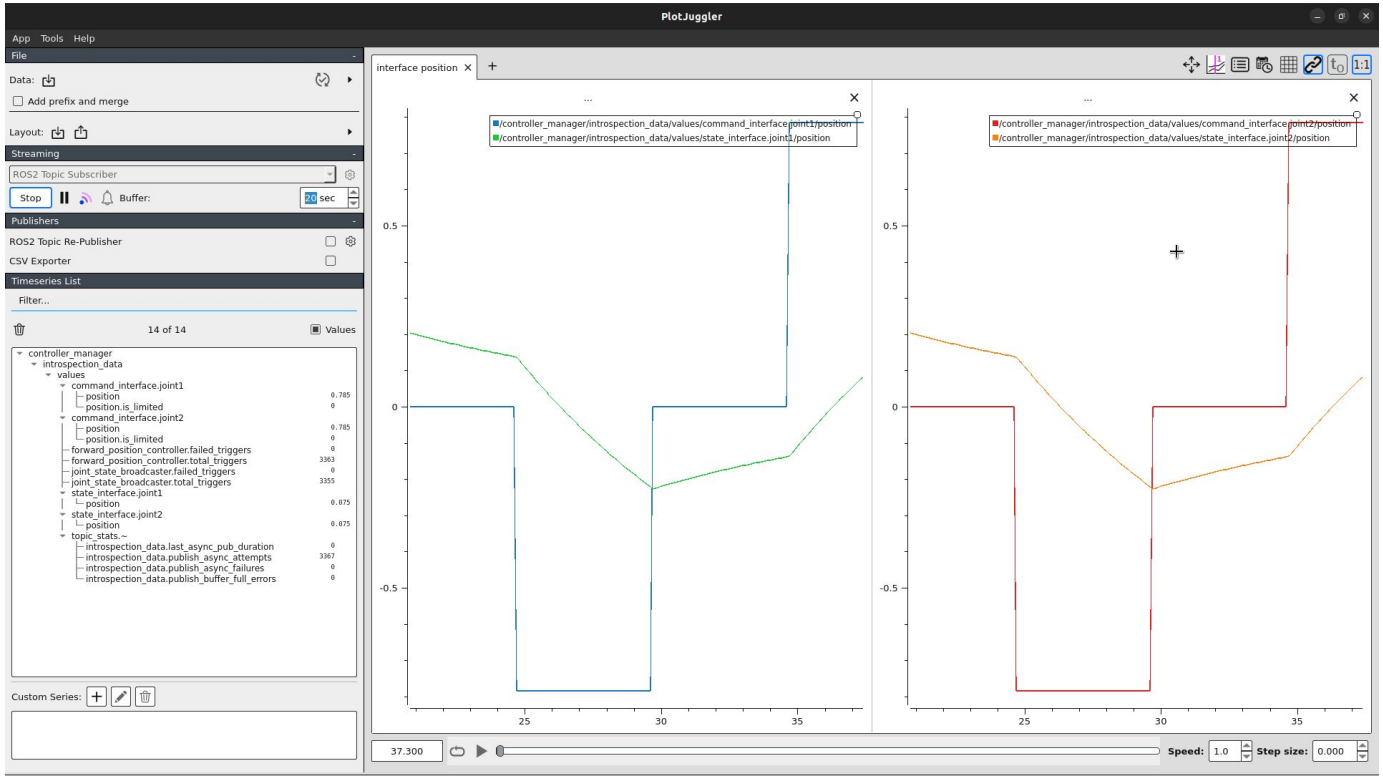
- ROS2 control provide macro for using pal_statisctics in controllers:
  - https://github.com/ros-controls/ros2_control/blob/jazzy/hardware_interface/include/hardware_interface/introspection.hpp#L47
  - https://github.com/ros-controls/ros2_control/blob/jazzy/controller_interface/src/controller_interface_base.cpp#L212
  - https://github.com/ros-controls/ros2_control/blob/jazzy/controller_interface/src/controller_interface_base.cpp#L356
- Example of Macros:

```
#define REGISTER_ROS2_CONTROL_INTROSPECTION(ID, ENTITY)                    \
REGISTER_ENTITY(                                                           \
    hardware_interface::DEFAULT_REGISTRY_KEY, get_name() + "." + ID, ENTITY, \
    &stats_registrations_, false);
```

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Pal_statistics in PlotJuggler

Version ≥3.10.11 : https://github.com/facontidavide/PlotJuggler/blob/main/CHANGELOG.rst



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Live demo

Available in nix ;)

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thank you